Notes: Kőszegi & Szeidl (QJE, 2013)
A Model of Focusing in Economic Choice

Contents

1 Big picture

- **Question.** How can economists model the fact that people pay disproportionate attention to attributes on which alternatives differ a lot, and what does this imply for welfare and intertemporal choice?
- **Core idea.** The model assumes that the decision maker focuses more on attributes with larger utility ranges across the current consideration set. Attributes that vary little across options receive less attention; attributes with large differences receive more attention.
- **Decision utility versus welfare.** The agent chooses by maximizing focus-weighted utility, but welfare is ordinary consumption utility. This separation lets the model speak about mistakes without assuming that attention changes experienced utility.
- **Main bias.** The paper identifies a **bias toward concentration**: people are too attracted to options whose advantages are concentrated in a small number of dimensions, even when their disadvantages are spread across many dimensions.
- **Balanced choices.** When advantages and disadvantages are similarly concentrated, focus does not distort the choice. The agent chooses as if she maximized true consumption utility.
- **Intertemporal application.** Treating utility at different dates as different attributes generates present bias in lifestyle decisions, but not necessarily in isolated single trade-offs. A current cost is salient because it is concentrated today, while future benefits may be dispersed across many periods.
- **Why this matters.** The theory explains why people respond more to concentrated incentives, such as money, than to dispersed incentives, such as health benefits. It also predicts overcommitment to concentrated goals.
- **Relation to hyperbolic discounting.** Focusing reproduces some field patterns usually attributed to hyperbolic discounting, but it also predicts different comparative statics: the concentration of costs and benefits matters independently of their timing.
- **Applications.** The framework can be used in product design, add-on pricing, annuity puzzles, social preferences, framing, work incentives, health behavior, and lifestyle commitment.
- **Takeaway.** The paper provides a portable behavioral tool: start from a classical utility model, define attributes, specify the consideration set, and replace ordinary utility maximization with focus-weighted utility maximization.

2 Incremental contribution of the paper

- **General formalization of focusing.** Earlier psychology and economics papers documented salience, framing, and focusing effects. This paper gives a general model that can be inserted into standard deterministic economic problems.
- **Endogenous focus weights from the choice environment.** Instead of assigning attention weights exogenously, the model derives focus weights from the range of utility differences across available options. This makes the model portable across domains.
- **A welfare benchmark for behavioral mistakes.** The paper explicitly distinguishes focus-weighted decision utility from consumption utility welfare. This allows a clean welfare analysis of when focusing causes mistakes.
- **Bias toward concentration as a unifying mechanism.** The paper shows that many distortions are not merely impatience, risk attitudes, or reference dependence. They arise because concentrated advantages draw too much attention relative to dispersed disadvantages.
- **Balanced-choice rationality.** The theory does not imply that people are always biased. It identifies a class of balanced trade-offs in which choices are welfare maximizing. This gives the model a sharp diagnostic test: the distribution of advantages and disadvantages across attributes determines bias.

- **A new account of present bias.** The paper shows that present bias can arise even without hyperbolic discounting if current costs are concentrated and future benefits are dispersed. This is an incremental contribution to intertemporal choice theory.
- **Comparative statics unavailable in hyperbolic discounting.** The model predicts that present bias changes with how benefits and costs are bundled, framed, or dispersed over time. Hyperbolic discounting predicts timing effects; focusing predicts concentration effects.
- **Ex ante welfare is not automatically first-best.** Although the ex ante perspective often better reflects welfare than ex post impulses, the paper shows that ex ante choices can themselves be distorted, producing overcommitment to concentrated goals or to beneficial activities.
- **Bridge from behavioral theory to applications.** The final section gives a recipe for translating deterministic classical models into focus-dependent models, making the contribution methodological as well as substantive.

3 Model objects and measurement

3.1 Choice set, consideration set, and attributes

- The model studies a finite set $C \subset \mathbb{R}^K$ of possible consumption vectors.
- Each dimension $k = 1, \ldots, K$ is an attribute. In an intertemporal application, different dates can be attributes. In a product setting, price and quality can be attributes. In a social setting, different people can be attributes.
- The set C is interpreted as a consideration set: the options that actually determine attention. It may be smaller than the technically feasible choice set.
- An option is written as $c = (c_1, \ldots, c_K)$, and the utility contribution of attribute k is $u_k(c_k)$.

Interpretation: The central empirical modeling choice in applying the paper is how to define attributes and the consideration set. Once those are specified, focus weights follow mechanically from the model.

3.2 Consumption utility and welfare

The classical welfare utility of option c is

$$U(c) = \sum_{k=1}^{K} u_k(c_k).$$

- $U(c)$ is the welfare criterion.
- The paper interprets $U(c)$ as experienced or consumption utility, not momentary decision utility.
- Focus changes perceived choice values at the moment of decision, but does not change how much the outcome contributes to welfare once consumed.

Interpretation: This assumption is what makes the model normative. If focus weights were themselves part of welfare, then overweighting salient attributes would not necessarily be a mistake. The paper instead treats focus as a decision distortion.

3.3 Focus-weighted utility

The decision maker chooses as if maximizing

$$\widetilde{U}(c, C) = \sum_{k=1}^{K} g_k u_k(c_k).$$

The focus weight on attribute k is

$$g_k = g(\Delta_k(C)),$$

where

$$\Delta_k(C) = \max_{c \in C} u_k(c_k) - \min_{c \in C} u_k(c_k),$$

and $g(\cdot)$ is strictly increasing.

- $\Delta_k(C)$ is the utility range of attribute k across the consideration set.
- Larger ranges generate larger focus weights.
- The model does not require the agent to consciously compute $\Delta_k(C)$; it is a representation of attention.

Interpretation: The model says that the salience of an attribute is comparative. A feature matters more psychologically when the options differ sharply on that feature.

3.4 Attributes are exogenous

- The paper treats the set of attributes as external to the formal model.
- This is both a strength and a limitation: the model is flexible, but applications require discipline in defining attributes.
- In time, the natural restriction is often to treat utilities at different dates as different attributes.
- In industrial organization, the natural attributes may be price, quality, fees, add-ons, and service characteristics.

Interpretation: Framing matters precisely because it may change the attribute representation. The same economic outcome can be perceived differently if a cost is described as one large payment versus many small payments.

3.5 The consideration set is exogenous

- Not every feasible option needs to affect focus.
- Focus is determined by options the agent actively considers as relevant comparisons.
- This allows the model to capture context effects: adding or removing comparison options can change focus weights and choices.

Interpretation: A product is not evaluated in isolation. The same product may appear different when compared to different alternatives because those alternatives change the attributes that stand out.

4 Models and theoretical specifications

4.1 Core model in one line

$$\text{Classical model: } \max_{c \in C} \sum_k u_k(c_k) \quad \longrightarrow \quad \text{Focusing model: } \max_{c \in C} \sum_k g(\Delta_k(C)) u_k(c_k).$$

Interpretation: A classical model becomes a focusing model by multiplying each attribute's utility by a weight that is increasing in the cross-option range of that attribute.

4.2 Two-option representation: pluses and minuses

For two options c^1 and c^2, define attributes in which c^1 is better than c^2 as pluses and attributes in which c^1 is worse as minuses. Let

$$P = \text{total pluses of } c^1 \text{ over } c^2, \qquad M = \text{total minuses of } c^1 \text{ relative to } c^2.$$

The welfare comparison depends on $P - M$. The focus-weighted comparison depends on the distribution of pluses and minuses across attributes.

Interpretation: The model is not only about how large the total gain and loss are. It is about how they are partitioned. One large benefit can receive more attention than many small costs whose total value is equally large or larger.

4.3 Proposition 1: bias toward concentration

- The agent is too prone to choose options with concentrated advantages and dispersed disadvantages.
- If the advantage of an option is concentrated on one or a few attributes, those attributes receive high focus weights.
- If the disadvantages are spread thinly across many attributes, each disadvantage receives a low focus weight.
- The agent can therefore choose an option whose true consumption utility is lower.

Interpretation: This result is the paper's main behavioral mechanism. It formalizes why immediate, vivid, or concentrated benefits can dominate many diffuse costs.

4.4 Proposition 2: increasing concentration

- Holding total benefits and costs fixed, concentrating the benefits makes the favored option more attractive.
- Holding totals fixed, spreading costs over more attributes makes the costs less psychologically powerful.
- This generates preferences for bundles with headline advantages and hidden or fragmented costs.

Interpretation: The proposition gives comparative statics. It predicts that firms, policymakers, or individuals can affect choices by repackaging the same payoffs across attributes.

4.5 Proposition 3: rationality in balanced trade-offs

- In balanced choices, advantages and disadvantages are similarly concentrated.
- Focus then reinforces the true ranking rather than distorting it.
- The decision maker maximizes consumption utility.

Interpretation: The model does not imply universal irrationality. It predicts mistakes in systematically identifiable environments: concentrated benefits versus dispersed costs, or concentrated costs versus dispersed benefits.

4.6 Proposition 4: more rationality in more balanced choices

- When a choice becomes more balanced in the relative number of plus and minus attributes, focus distortions fall.
- The paper uses this idea in social examples such as tax evasion: a person may ignore many small harms to others in a single case, but recognize the costs when many similar decisions are considered together.

Interpretation: Aggregating decisions can make dispersed effects more salient. This is why people may think differently about one person's tax evasion and widespread tax evasion, even if the per-case cost-benefit structure is similar.

4.7 Framing

- Framing can change how payoffs are allocated across perceived attributes.
- A cost described as "$39 per month" may receive less focus than the same cost described as "$936 in total."
- Firms may exploit this by splitting prices into small add-on fees or emphasizing a product's concentrated advantages.

Interpretation: Framing works because it changes concentration. The model provides a formal reason why "pennies-a-day" marketing can increase demand.

4.8 Relationship with diminishing sensitivity

- Diminishing sensitivity says that marginal sensitivity to an outcome is lower farther from the reference point.
- Focusing says that attributes with larger cross-option differences receive more attention.
- These forces can be orthogonal, offsetting, or mutually reinforcing.

Interpretation: Focusing is not simply prospect theory. It is a separate attention channel that can explain patterns that diminishing sensitivity alone cannot.

5 Intertemporal choice application

5.1 Utility dates as attributes

- In the time-choice application, each date's utility is a separate attribute.

- The agent in period t considers lifetime consumption profiles made possible by current choices and beliefs about future behavior.
- Focus weights are determined by the variation in per-date utilities across the considered profiles.

Interpretation: Time inconsistency can arise even without changing discount factors. It arises because the salience of costs and benefits changes when decisions are viewed narrowly or broadly.

5.2 Single balanced choices

- A single intertemporal trade-off, such as a massage today versus a larger massage next week, is often balanced.
- Both options have one main advantage and one main disadvantage.
- The model therefore predicts no present bias or time inconsistency in this kind of single choice.

Interpretation: This is a sharp difference from hyperbolic discounting, which predicts present bias even in isolated single trade-offs.

5.3 Lifestyle choices

- A lifestyle choice involves repeated decisions whose effects accumulate.
- A current temptation often has a concentrated benefit or avoids a concentrated cost today.
- The long-run health, wealth, or skill benefits are often spread across many future periods.
- The agent therefore underweights future consequences ex post and exhibits present bias.

Interpretation: The model explains why present bias is strongest in domains like exercise, smoking, saving, or diet: the immediate payoff is concentrated, while the future consequences are diffuse.

5.4 A stylized investment problem

The paper studies an activity that requires effort of size B and yields future benefits. In the ex post decision, the current cost is concentrated in one period, while benefits are spread over T_b future periods. The agent invests when the focus-weighted benefit exceeds the focus-weighted cost. A useful threshold representation is

$$A^*_{post} = h(T_b),$$

where A measures the average return to the activity and $h(\cdot)$ is increasing.

In the ex ante decision, the agent considers multiple effort decisions, so benefits can look more concentrated relative to the full plan. The threshold becomes

$$A^*_{ante} = h\left(\frac{T_b}{T_i}\right),$$

where T_i is the number of investment periods.

Interpretation: The longer the future benefits are dispersed, the more present-biased the ex post choice is. The more decisions are bundled ex ante, the more the agent may want to commit.

5.5 Proposition 5 and Corollary 1

- Ex post investment is too low when current costs are concentrated and future benefits are dispersed.
- Ex ante commitment can reduce this problem because repeated efforts and repeated benefits are compared more symmetrically.
- But ex ante commitment can also be excessive when a concentrated future goal receives too much focus.
- A person can overcommit to activities that are beneficial overall if the total benefit is salient relative to the repeated costs.

Interpretation: The ex ante self is not always a welfare benchmark. Focus can distort long-run planning as well as short-run temptation.

6 Theoretical design and identification logic

6.1 What the paper identifies

- The paper identifies a mechanism: the concentration of utility differences across attributes affects attention and choice.
- It identifies classes of choices in which the mechanism does and does not create welfare distortions.
- It identifies intertemporal settings in which focusing mimics present bias and settings in which it does not.

Interpretation: In a theoretical paper, "identification" means the logic that isolates the causal mechanism in the model. The mechanism is attribute-range-based attention, not impatience or changing tastes.

6.2 Why the model is disciplined

- Focus weights depend only on objects already present in the decision problem: options, attributes, and utility functions.
- Welfare is fixed as consumption utility, so the model has a clear benchmark for mistakes.
- The model produces both distortion and no-distortion cases, making it more falsifiable than a model that can rationalize any choice.

Interpretation: The model's strength is that it generates sign restrictions and comparative statics from a small number of assumptions.

6.3 Main limitations

- Attributes are not derived endogenously.
- The consideration set is not derived endogenously.
- The model is mainly deterministic, and extensions to uncertainty require additional assumptions.
- Focus weights depend on the scaling of the utility function unless the focus-weight function is adjusted accordingly.
- The model does not replace hyperbolic discounting; it explains some but not all evidence about time inconsistency.

Interpretation: These limitations are important for empirical applications. A researcher must justify the attributes, the relevant comparison set, and the utility scale.

7 Section-by-section study guide

7.1 Section I: Introduction

- Introduces the hypothesis that people focus on attributes with larger differences across options.
- Provides motivating examples: California climate, unhealthy food prices versus health costs, intertemporal self-control.
- Sets up the distinction between focusing and hyperbolic discounting.

7.2 Section II: A theory of focusing

- Defines focus-weighted utility.
- States the focus-weight assumption and welfare assumption.
- Discusses modeling issues: exogenous attributes, exogenous consideration set, framing, reference dependence, and utility scaling.

7.3 Section III: General implications

- Establishes bias toward concentration.
- Shows rationality in balanced choices.
- Discusses framing and relationship with diminishing sensitivity.

7.4 Section IV: Intertemporal choice

- Applies the model to utility streams over time.
- Shows no present bias in single balanced choices.
- Shows present bias in lifestyle choices with concentrated costs and dispersed future benefits.
- Derives commitment, overcommitment, and future-bias predictions.

7.5 Section V: Other applications and conclusion

- Provides a recipe for applying focus-dependent utility to other domains.
- Discusses product design, add-ons, annuities, social preferences, and work incentives.
- Emphasizes that the theory is a general modeling framework, not only a theory of intertemporal choice.

8 Tables and figures

8.1 Table I: A Summary of the Main Predictions of Focusing and Hyperbolic Discounting

Read:

- The table contrasts focusing with hyperbolic discounting along two dimensions: behavior and welfare.
- For single choices, focusing predicts no time inconsistency or present bias, while hyperbolic discounting predicts both.
- For lifestyle choices, both models predict time inconsistency and present bias.
- Focusing uniquely predicts higher responsiveness to concentrated than to dispersed incentives.
- Focusing uniquely predicts a stronger tendency to make long-term commitments than short-term commitments.
- On welfare, focusing predicts overcommitment to concentrated goals and overcommitment to an overall beneficial lifestyle; hyperbolic discounting does not.

Detailed interpretation:

- The single-choice row is crucial because it tells us where focusing diverges from quasi-hyperbolic discounting. If the choice is balanced, focus reinforces the correct trade-off.
- The lifestyle-choice row explains why focusing can mimic present bias in the field: many empirically studied self-control problems involve repeated decisions and diffuse cumulative consequences.
- The incentives row is the main comparative-statics innovation. Monetary rewards or fines can be powerful because they are concentrated, while health consequences or small future losses can be weak because they are dispersed.
- The commitment rows show why welfare analysis is subtle. Ex ante commitment is often helpful, but the ex ante self may also overcommit when a future target is salient.

Economic message: The table is the paper's clearest statement of incremental contribution. Focusing can explain many of the same lifestyle patterns as hyperbolic discounting, but it also predicts when those patterns should weaken, reverse, or appear in non-time-preference domains.

TABLE I

A Summary of the Main Predictions of Focusing and Hyperbolic Discounting

	Focusing	Hyperbolic disc.
Behavior		
Time inconsistency in single choice	no	yes
Time inconsistency in lifestyle choices	yes	yes
Higher responsiveness to concentrated than to dispersed incentives	yes	no
More likely to make long-term than short-term commitments	yes	no
Welfare		
Present bias in single choice	no	yes
Present bias in lifestyle choices	yes	yes
Ex ante perspective generally better for welfare	yes	yes
Overcommitment to concentrated goals	yes	no
Overcommitment to overall beneficial lifestyle	yes	no

Table I: A Summary of the Main Predictions of Focusing and Hyperbolic Discounting

8.2 Paired theoretical result blocks: model primitives and general results

Result block 1: Focus-weighted utility

Classical welfare is $U(c) = \sum_k u_k(c_k)$. Decision utility is $\widetilde{U}(c, C) = \sum_k g(\Delta_k(C))u_k(c_k)$. Focus is stronger on attributes with larger ranges across the consideration set.

Result block 2: Welfare benchmark

Welfare is consumption utility, not focus-weighted utility. This lets the paper identify attention-driven mistakes and compare choices to a stable normative benchmark.

Result block 3: Bias toward concentration

A concentrated advantage can be overweighted relative to many small disadvantages. This generates excessive choice of options with salient headline benefits and hidden diffuse costs.

Result block 4: Balanced-choice rationality

When advantages and disadvantages are equally concentrated, focus does not distort the ranking. The agent chooses as if maximizing consumption utility.

Result block 5: Framing

Changing the representation of the same outcomes can change concentration across attributes. This gives a formal mechanism for price partitioning, pennies-a-day framing, and add-on pricing.

Result block 6: Diminishing sensitivity

Focusing and diminishing sensitivity are distinct forces. Depending on the representation, they can reinforce, offset, or be orthogonal to each other.

8.3 Paired theoretical result blocks: intertemporal choice

Result block 7: No present bias in single balanced choice

A single trade-off between one current benefit and one future benefit can be balanced. Focusing then predicts no time inconsistency, unlike hyperbolic discounting.

Result block 8: Present bias in lifestyle choices

Repeated lifestyle decisions create concentrated current costs or benefits and dispersed future consequences. This creates present bias and time inconsistency through focusing.

Result block 9: Commitment

Ex ante commitment can help when it bundles future benefits and repeated decisions into a more balanced comparison. This predicts stronger demand for long-run than short-run commitments.

Result block 10: Overcommitment

Ex ante choice can be distorted too. A concentrated goal, bonus, or achievement can draw too much focus, leading to excessive commitment to work, exercise, or other activities.

9 Result map

- **Assumption 1.** Focus weights increase with the attribute's utility range across the consideration set.
- **Assumption 2.** Welfare is consumption utility, not focus-weighted utility.
- **Proposition 1.** The agent is biased toward options with concentrated advantages and dispersed disadvantages.
- **Proposition 2.** Increasing concentration of an option's advantages makes it more attractive, holding total utility differences fixed.
- **Proposition 3.** In balanced trade-offs, the agent maximizes consumption utility.
- **Proposition 4.** Choices become more rational as pluses and minuses become more balanced in concentration.

- **Proposition 5.** In intertemporal investment, focusing generates present bias and time inconsistency when future benefits are dispersed relative to current costs.
- **Corollary 1.** The model predicts specific comparative statics for lifestyle choices: more bias when future consequences are more dispersed and more commitment when future benefits become more salient.
- **Table I.** Summarizes the difference between focusing and hyperbolic discounting.
- **Section V.** Gives a recipe for applying the model outside intertemporal choice.

10 Short summary

- The paper develops a general model of focus-dependent choice.
- The agent's attention is drawn to attributes on which options differ more.
- The agent maximizes focus-weighted utility, but welfare is ordinary consumption utility.
- The main distortion is a bias toward concentration.
- Concentrated benefits can dominate dispersed costs even when total costs are larger.
- Balanced trade-offs generate no distortion.
- Framing matters because it can change how payoffs are distributed across perceived attributes.
- In intertemporal choice, focusing can generate present bias in lifestyle choices without assuming hyperbolic discounting.
- The model predicts no present bias in isolated balanced choices.
- The model predicts that concentrated incentives are more effective than dispersed incentives.
- Ex ante commitment can improve decisions, but can also generate overcommitment.
- The model helps explain responses to monetary incentives, health behavior, add-on pricing, annuity choices, and work incentives.
- The paper's contribution is methodological: it provides a recipe for adding focus to classical deterministic models.

11 Conclusion

11.1 Core conclusion

Kőszegi and Szeidl provide a general theory of focusing in economic choice. The model's key claim is that people overweight attributes along which available options differ substantially. This simple attentional rule generates systematic and welfare-relevant distortions, especially when an option's benefits are concentrated and its costs are dispersed.

11.2 Economic intuition

The intuition is that attention is comparative. A single large gain stands out more than many small losses. A single current cost of effort stands out more than many small future benefits. A single future goal can stand out more than many repeated sacrifices. The model converts this intuition into a formal utility representation in which focus weights are increasing functions of cross-option utility ranges.

11.3 Incremental contribution relative to prior behavioral models

The paper differs from hyperbolic discounting because it does not make impatience the primitive. It differs from prospect theory because it is not primarily about reference points and diminishing sensitivity. It differs from informal focusing-illusion evidence because it provides a portable formal model. The result is a theory that can explain present-bias-like behavior in lifestyle choices while also predicting no present bias in balanced single choices, a pattern that hyperbolic discounting cannot generate by itself.

11.4 Welfare implications

The welfare implications are subtle. In many lifestyle choices, the ex ante perspective is better because the ex post self underweights dispersed future benefits. But ex ante choices can also be distorted if the benefits of commitment are concentrated. Thus the paper does not simply say that commitment is always good. It says that the welfare value of commitment depends on whether the commitment reframes the decision in a more balanced way or instead overfocuses attention on a concentrated goal.

11.5 Policy and market-design implications

The model implies that policies and contracts can change behavior by changing concentration rather than by changing total incentives. Concentrated incentives, such as cash rewards, may be more effective than dispersed health information. Lump-sum costs may deter behavior more than installment costs. Product sellers may exploit focus by emphasizing concentrated product benefits and splitting costs into small fees. Regulators may need to pay attention not only to prices and information, but also to the attribute representation of transactions.

11.6 Limitations

The theory leaves open how attributes and consideration sets are formed. It also requires discipline in scaling utility and choosing the focus function. Because the main model is deterministic, applications to uncertainty require additional structure. Finally, the model is not a complete theory of intertemporal choice; the authors explicitly note that hyperbolic discounting or related models may still be needed to explain single-choice present bias.

11.7 Takeaway

The enduring message is that economic choices depend not only on what options deliver, but also on how the advantages and disadvantages of those options are distributed across the attributes people attend to. A researcher who wants to model behavior in settings with framing, salience, attention, or lifestyle temptation can begin with a standard utility model and then ask which attributes will draw focus because the options differ most strongly on them.